SORCE CODE:

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#include <stdio.h>
#include <math.h>
#include <float.h>

// Function to find the closest pair distance
double closestPair(int points[][2], int n)
{
    double minDist = DBL_MAX;

    for (int i = 0; i < n; i++)
    {
        // Compare only with the remaining points
        for (int j = i + 1; j < n; j++)
        {
            double dx = points[i][0] - points[j][0];
            double dy = points[i][1] - points[j][1];

            double dist = sqrt(dx * dx + dy * dy);

            if (dist < minDist)
            {
                minDist = dist;
            }
        }
    }

    return minDist;
}

int main()
{
    // Sample points
    int points[][2] = {
        {2, 3},
        {12, 30},
        {40, 50},
        {5, 1},
        {12, 10},
        {3, 4}
    };

    int n = sizeof(points) / sizeof(points[0]);

    double result = closestPair(points, n);

    printf("Minimum Distance = %.2lf\n", result);
```

```
    return 0;
}
```